Sujal Thapa

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Experience

Accelerated Komputing — Penn Ave, Pittsburgh
OJT — *2025-Present*

Accelerated Komputing — Penn Ave, Pittsburgh
Remote Internship — *2025, October*

MeroSpace, SOMAES — Puclhowk Campus, Lalitpur
Editor — *2023 - 2025*

Pulchowkians' Fitness Hub — Lalitpur
Marketing Lead — *2023 - 2024*

Education

Institute of Engineering, Pulchowk Campus — Lalitpur, Nepal
Bachelors in Mechanical Engineering — *2022- 2026*

- **Current Aggregate Percentage:** 77.3% (Excluding final semester)
- **Core Courses:** Engineering Thermodynamics, Fluid Mechanics, Heat and Mass Transfer, Applied Mechanics (Statics and Dynamics), Strength of Materials, Machine Design, Theory of Machines, Mechanical Vibration, Metrology and Instrumentation, Manufacturing Processes, Material Science, Control Systems, IC Engines, Refrigeration and Air Conditioning, Computer-Aided Design (CAD), Engineering Drawing and Design, Industrial Engineering, Engineering Mathematics (I, II, III)

Projects

Surrogate-Based Modeling for Aeroelastic Simulation of a Hypersonic Intake

Final Year Project — *2025 April - 2026 March*

- Worked on creating a gradient-informed gaussian process surrogate model for modeling TPR response of a hypersonic intake. This included:
    - Latin Hypercube Sampling for deformation states.
    - Baseline simulations for varying deformations of the intake.
    - Computationally efficient sensitivity simulations for gradient information acquisition.
    - Gradient-leveraged surrogate model construction.
    - Reduced order modeling for transient reconstruction from static models.
- *Skills:* CFD, Sensitivity Simulations, Supervised Learning Methodologies, Gaussian Surrogate Models, Gradient-Enhanced Surrogates.
- *Tools:* OpenFOAM, Python, ANSYS ICEM CFD, Python libraries like SMT, numpy, pandas, and matplotlib.

Stress Distribution Prediction of Cellular Beams Using RegDGCNN

AnK Internship Project *2025 October*

- Worked on implementing Regression on Dynamic Graph Convolutional Neural Network (RegDGCNN) to predict stress distribution on cellular beams of varying hole diameter. The project involved:
    * Geometry preparation and baseline simulations of cellular beams.
    * Data extraction and cleaning in H5 file format.
    * RegDGCNN architecture development.
    * Validation and testing.
- *Skills:* FEM, ANSYS Simulations, Neural Networks, Graph-Neural Networks, RegDGCNN
- *Tools:* Python, ANSYS, Python libraries: SMT, numpy, pandas, and matplotlib.

Efficient Mesh Morphing Strategy for OpenFOAM Cases using ISDW

2025 June - September

- Worked on writing a python script that used ISDW (Inverse Square Distance Weighting) algorithm to efficiently morph meh without having to manually perturb the geometry and re-mesh. The project involved::
    * Extracting geometry files from OpenFOAM case folder in python.
    * Extracting relevant points from the patches.
    * Applying modeled deformation to the appropriate patches.
    * Applying ISDW to deform all non-involved internal points of the mesh.
    * Updating the mesh files according to the morphed configuration.
- *Skills:* Structural modeling, file handling in python, OpenFOAM fundamentals.
- *Tools:* OpenFOAM, Python, ANSYS, Python file handling libraries.

Gaussian Process Regression for a Hypersonic Wedge Flow Simulation

Benchmark Study *2025 June - August*

- A preliminary project to observe the total reduction in wall-clock time when constructing a surrogate that used gradient information from sensitivity simulations on top of baseline simulations. The project involved:
    - Design of experiment using Latin Hypercube Sampling.
    - Determination of step-size for sensitivity simulations.
    - Determination of step-size for sensitivity simulations.
    - Exploratory data analysis.
    - Construction and comparison of OK and GEK models.
- *Skills:* Hypersonic simulations, Kriging and GEK surrogate construction, Warm start optimization in CFD
- *Tools:* OpenFOAM, Python, ANSYS, Python libraries like SMT, numpy, pandas, and matplotlib.

Key Skills

Statistical Learning: Regression Models, Kriging Modalities, Gradient-Enhanced Kriging Models.

Programming Language: Python, MATLAB, C (in order of confidence.)

Computational Tools: OpenFOAM, ParaView, FreeCAD (All open-source software.)

Transferrable Skills: Proficiency at crafting and editing documents and articles, Collaboration & interpersonal skills, Communication and media literacy

Summary

A mechanical engineering student from Pulchowk Campus with a strong interest in efficient computational means to solve pratical engineering problems. Experienced in leadership and an overall collaborative environment within Pulchowk Campus. Familiar with Python for data analysis and surrogate modeling libraries such as SMT, NumPy, pyDOE, and matplotlib.